



Medreach

RIGHT HOSPITAL · RIGHT NOW

The right hospital, right now, **confirmed before you leave home**

AI-powered emergency coordination for Madhya Pradesh: understand the emergency, find a hospital that can treat it now, get it to accept, and guide the patient there.

MPOnline Hackathon 2026

Challenge 5 · Healthcare

Team Leader: Prakhar Porwal

Live demo: medreach-2.onrender.com · Code: github.com/prakharporwal518-collab/Medreach



The nearest hospital is not always the right one

11 pm, Bhopal. A 62-year-old man has chest pain. His family rushes to the nearest hospital, where there is no cardiologist on duty and no free ICU bed. They are sent across the city, and the golden hour is lost.

1,68,491

road-accident deaths in India, 2022

Source: MoRTH, "Road Accidents in India 2022". Heart attacks, strokes, burns and obstetric emergencies add many more time-critical cases.



Nearest ≠ capable

Maps show the closest hospital, not whether the specialist is on duty, the ICU is free or the CT scanner works now.



No trusted live status

Bed and doctor availability is unpublished or outdated, with no owner and no time stamp.



No acceptance first

Families arrive unannounced; hospitals cannot prepare and refusals happen at the gate.

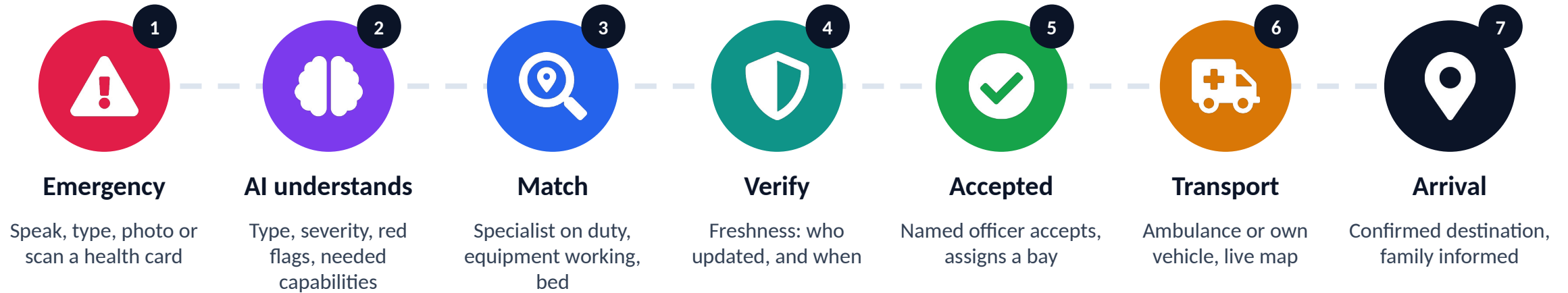


Panic & living alone

Hard to type, call hospitals, arrange transport and inform family, especially for elderly people alone.

One flow from emergency to confirmed bed

An emergency coordination platform, not a diagnosis app. Every step of our flowchart is implemented and live.



**Hindi · English ·
Hinglish**

Voice-first, built for panic

**Only capable
hospitals**

Deterministic, explainable
matching

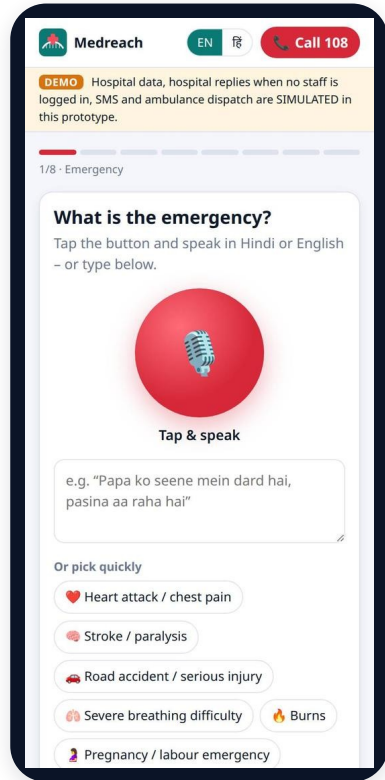
**Accepted before
travel**

No refusals at the gate

60 s auto-fallback

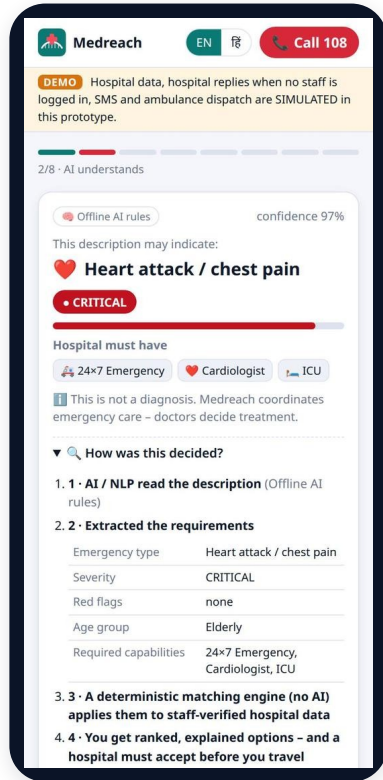
No answer → next hospital asked

For citizens: speak, choose, get accepted



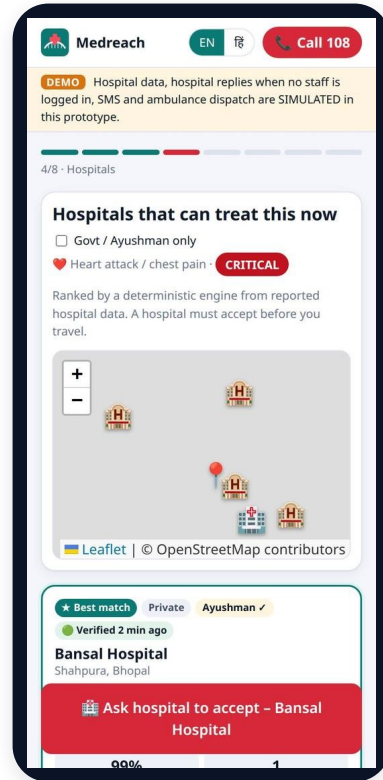
1 • Speak or type

Hindi or English voice, quick picks, photo and health-card scan



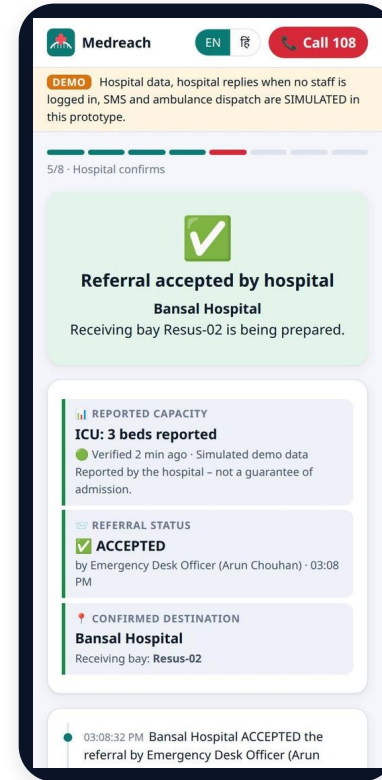
2 • AI understands

“May indicate heart attack • CRITICAL • needs cardiologist + ICU”



3 • Capable hospitals

Ranked from staff-verified data, with “Why this hospital?”



4 • Accepted

Named officer accepts, receiving bay Resus-02 ready

What the citizen gets



Voice in Hindi & English



“How was this decided?” explained



Call 108 always one tap away



Works offline & on weak networks

Capability-first matching you can trust



Must-have capabilities

A hospital qualifies only if the service exists, the specialist is on duty and the machine works now.



Freshness on every figure

Every number shows who updated it and when. Stale data is never shown as confirmed.



Safe, explainable AI

AI only understands; a deterministic engine decides. Severity = max(AI, rules), so AI can never lower urgency.



Verified < 15 min



15–60 min: may have changed

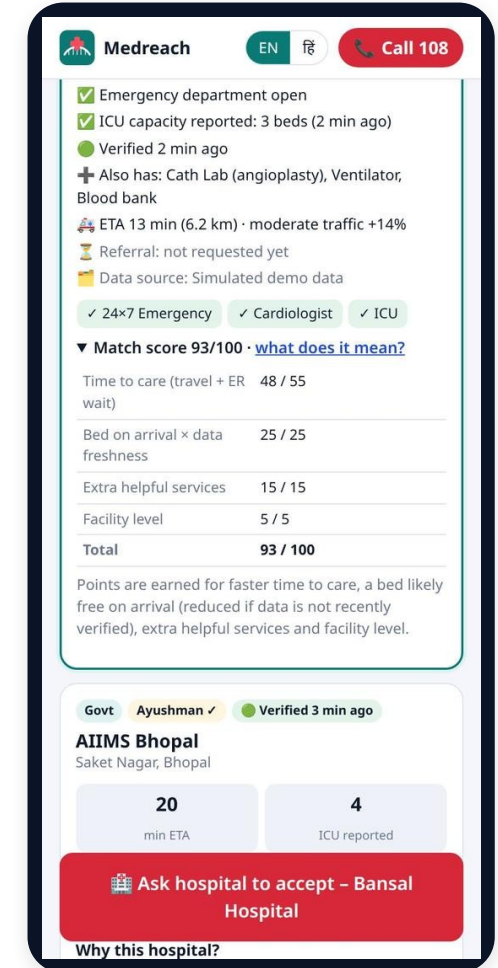


Unverified: ranked lower

Why Bansal Hospital scored 93 / 100

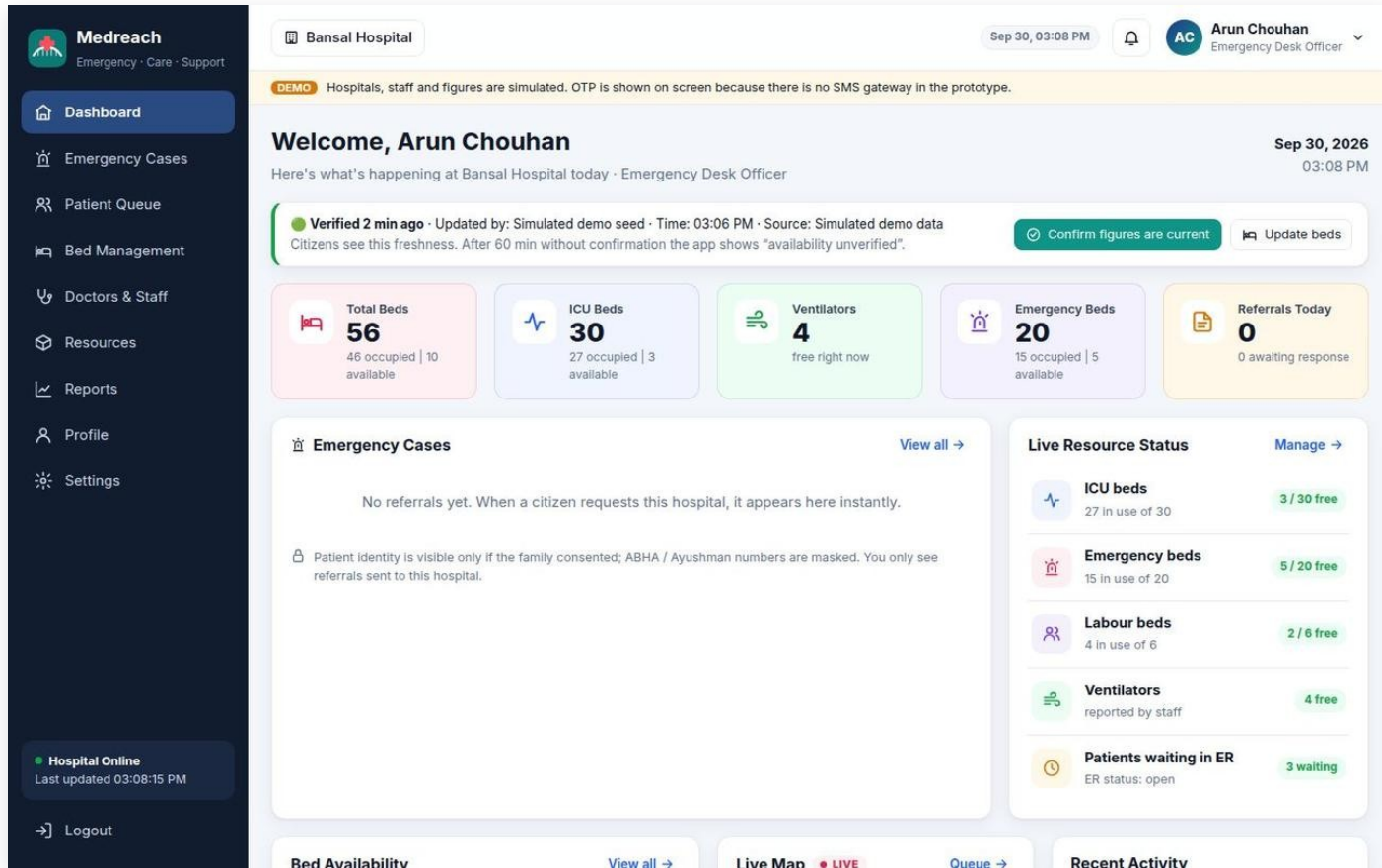


Score parts always add up to the total (tested). Predictive analytics: traffic-aware ETA, Poisson bed-on-arrival, ER wait.



“Why this hospital?” in the app

For hospitals: verified status, accept in one tap



Hospital dashboard: beds, ICU, ventilators and referrals, each with “updated by / time / source”



AI pre-arrival SBAR note

Situation, Background, Assessment, Recommendation before the patient arrives.



Accept + receiving bay

Named officer accepts and assigns e.g. Resus-02; citizen sees it instantly.



60-second auto-fallback

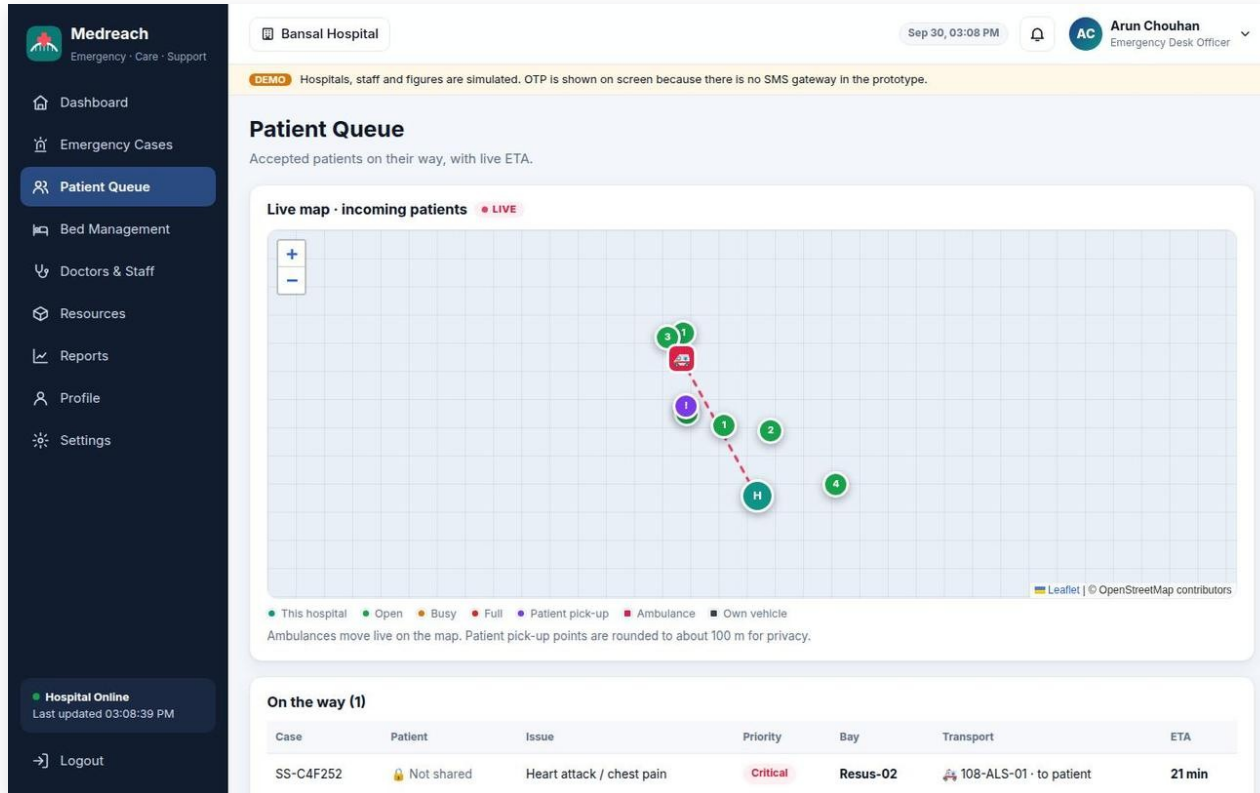
Decline or no answer → next suitable hospital is asked automatically.



Staff-only, audited

Hospital ID + Staff ID + OTP; 4 roles; every action in the audit log.

Live tracking, family link and “I’m alone” mode



Hospital Patient Queue: incoming patient, moving ambulance and live ETA



Ambulance or own vehicle



Free maps, no API keys



Family link: no medical data

“I’m alone” mode

One tap does every step for someone who cannot make calls.



- 1 Location shared
- 2 Trusted contact alerted with a tracking link
- 3 Hospital request sent with “patient is ALONE”
- 4 Hospital accepted, bay prepared
- 5 Ambulance requested

All 7 suggested technologies, working in the product



Generative AI

Claude turns messy text into requirements and writes the SBAR note

`server/llm.js`



Voice Bots

Speak in Hindi / English; key steps are read aloud

`public/js/voice.js`



Computer Vision

Photo of injury → visible findings that can raise urgency

`public/js/vision.js`



OCR

Ayushman / ABHA card read on the phone, shared only with consent

`public/js/ocr.js`



Predictive Analytics

Traffic ETA, bed-on-arrival odds, ER wait, 8-hour forecast

`shared/predict.js`



Conversational AI

One follow-up question at a time; severity updates live

`shared/triage.js`



Mobile App

Installable PWA, bilingual, works offline

`public/sw.js`



Plus: live maps

Citizen + hospital maps on free OpenStreetMap

`public/js/livemap.js`

Design principle: AI understands the citizen; a deterministic, explainable engine chooses the hospital from staff-verified data. Offline? The rules engine and PWA keep working.

Impact for people, hospitals and the state



Social

Right care the first time; equal access for rural, elderly and Hindi-speaking citizens



Economic

No repeat trips or wasted visits; better use of ICU beds and specialists



Governance

Live, audited view of emergency capacity for planning beds and ambulances



Educational

Safe first-aid steps, “not a diagnosis” clarity, awareness of Ayushman, ABHA, 108



User

One button, voice, bilingual, clear status and a tracking link for family



Health system

Fewer surprise arrivals, less ER crowding, fewer inter-hospital transfers

Privacy & security by design



Consent before sharing patient details



ABHA / Ayushman IDs masked



Staff OTP login and roles



Audit log of every action



Details deleted 24 h after hand-over



Aligned with the DPDP Act 2023

Pilot KPIs: time to definitive care · secondary transfers avoided · referral acceptance time · data freshness

Built today, ready to pilot in Bhopal

Thank you



Prototype

Done

Both portals live, AI triage, matching, acceptance, live maps, 51 tests



Pilot • Bhopal

~3 months

Onboard hospitals, nodal officers, ER desk training, clinical review



Integrate

~3-6 months

Health Facility Registry, hospital HMIS, 108 control room, ABDM, SMS



Scale • MP

~6-12 months

District-by-district rollout, state view, IVR for feature phones

Why Medreach should be selected

Real, urgent problem

Life-or-death gap for MP citizens

Working product

Deployed, 7/7 technologies, 51 tests

Safe & honest

Explainable, private, data labelled

Ready to scale

HFR, HMIS, 108, ABDM integration

Live demo: medreach-2.onrender.com Code: github.com/prakharpurwal518-collab/Medreach Hospital login: Bansal Hospital • BANSAL-ED01